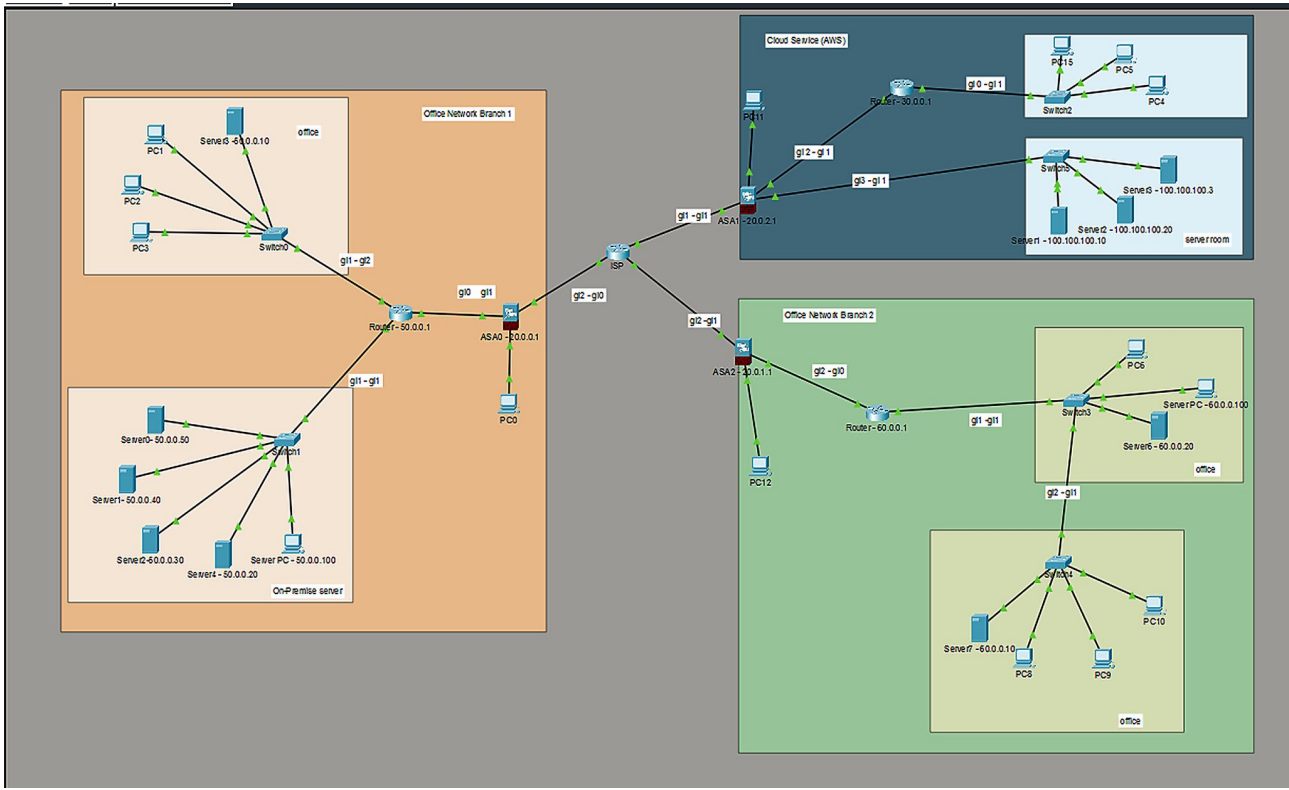
The Topology

Branch 1 and Branch 2 each sit behind their own ASA. The cloud servers sit behind a third one. A single ISP router joins all three.



The rules

Both branches can reach the cloud servers

Each branch can reach the other branch’s published servers

The Server PC in each branch gets a VPN to the other branch

Normal PCs cannot reach the other branch’s PCs

Addressing

Branch 1 LANs: 192.168.50.0/24 and 192.168.100.0/24

Branch 2 VLANs: 200.200.100.0/24 (VLAN 10) and 200.200.200.0/24 (VLAN 20)

Router to ASA links: 50.0.0.0/24 and 60.0.0.0/24

ASA to ISP: three /30 links (20.0.0.0, 20.0.1.0, 20.0.2.0)

Cloud servers: 100.100.100.0/24

**These are lab addresses, not RFC 1918.

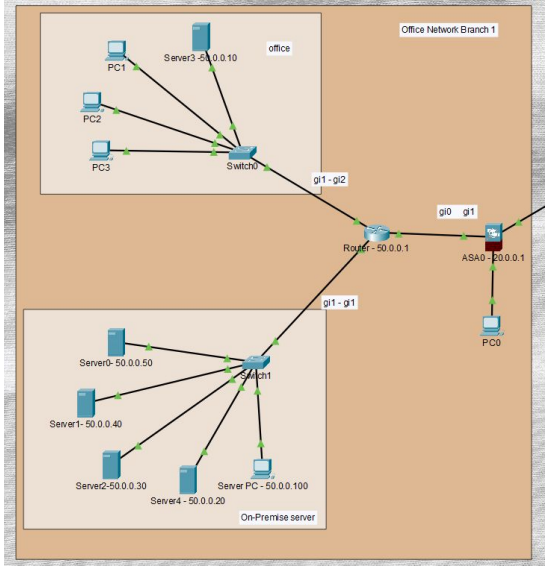
Branch routers

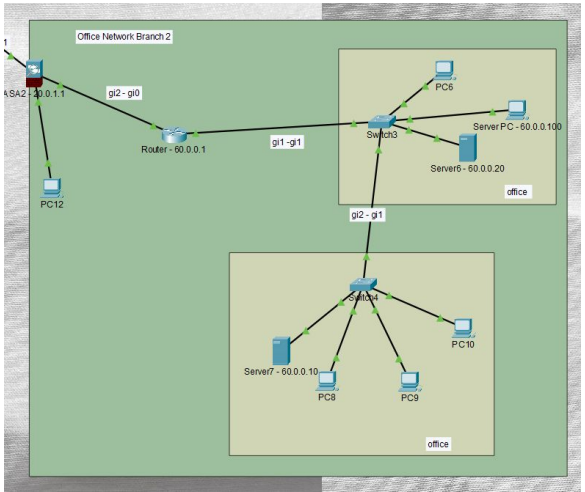
DHCP on every LAN, one pool each

Branch 2 uses router-on-a-stick for VLAN 10 and VLAN 20

PAT out of the router, plus static NAT for each server

Branch 1 has 6 static mappings, Branch 2 has 3





The ASAs

PAT for the whole inside network

Identity NAT for each published server, so it keeps its address

Outside ACLs that open only FTP, HTTPS and RDP to the right servers

The cloud ASA only accepts traffic from 20.0.0.1, 20.0.1.1, 50.0.0.0/24 and 60.0.0.0/24

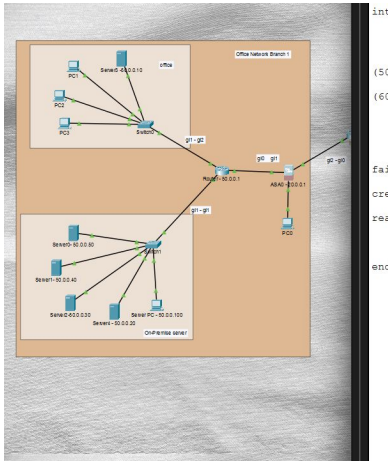
IPsec VPN

Tunnel between ASA0 and ASA2

IKEv1 with AES, a pre-shared key and DH group 5

ESP with AES-256 and SHA

A crypto ACL that matches only the traffic meant for the tunnel




```
interface: outside
    Crypto map tag: CMAP, seq num: 10, local addr 20.0.0.1

      permit ip object LOCALLAN object REMOTELAN
      local  ident (addr/mask/prot/port):
(50.0.0.0/255.255.255.0/0/0)
      remote  ident (addr/mask/prot/port):
(60.0.0.0/255.255.255.0/0/0)
      current_peer 20.0.1.1
      #pkts encaps: 5295, #pkts encrypt: 5295, #pkts digest: 0
      #pkts decaps: 9194, #pkts decrypt: 9194, #pkts verify: 0
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts comp failed: 0, #pkts decomp
failed: 0
      #pre-frag successes: 0, #pre-frag failures: 0, #fragments
created: 0
      #PMTUs sent: 0, #PMTUs rcvd: 0, #decapsulated frgs needing
reassembly: 0
      #send errors 1, #recv errors 0

      local crypto endpt.: 20.0.0.1/0, remote crypto
endpt.:20.0.1.1/0
      path mtu 1500, ip mtu, ipsec overhead 78,  media mtu 1500
      current outbound spi: 0x7B321C38(2066881592)
      current inbound spi: 0x7EA56673(2066881592)

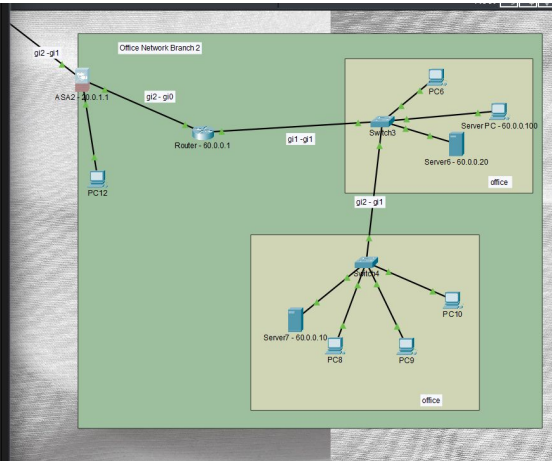
    inbound esp sas:
     spi: 0x7EA56673(2124768883)
        transform: esp-aes 256 esp-sha-hmac no compression
        in use settings ={L2L, Tunnel, }
        slot: 0, conn id: 2002, crypto map: CMAP
        sa timing: remaining key lifetime (k/sec): (4525504/83)
        IV size: 16 bytes
        replay detection support: N
        Anti replay bitmap:
         0x00000000 0x0000001F
    outbound esp sas:
```

```
interface: OUTSIDE
    Crypto map tag: CMAP, seq num: 10, local addr 20.0.1.1

      permit ip object LOCAL_LAN object REMOTE_LAN
      local  ident (addr/mask/prot/port):
(60.0.0.0/255.255.255.0/0/0)
      remote  ident (addr/mask/prot/port):
(50.0.0.0/255.255.255.0/0/0)
      current_peer 20.0.0.1
      #pkts encaps: 29, #pkts encrypt: 29, #pkts digest: 0
      #pkts decaps: 21, #pkts decrypt: 21, #pkts verify: 0
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts comp failed: 0, #pkts decomp
failed: 0
      #pre-frag successes: 0, #pre-frag failures: 0, #fragments
created: 0
      #PMTUs sent: 0, #PMTUs rcvd: 0, #decapsulated frgs needing
reassembly: 0
      #send errors 1, #recv errors 0

      local crypto endpt.: 20.0.1.1/0, remote crypto
endpt.:20.0.0.1/0
      path mtu 1500, ip mtu, ipsec overhead 78,  media mtu 1500
      current outbound spi: 0x0531BBF6(87145462)
      current inbound spi: 0xF96FC702(87145462)

    inbound esp sas:
     spi: 0xF96FC702(4184852226)
        transform: esp-aes 256 esp-sha-hmac no compression
        in use settings ={L2L, Tunnel, }
        slot: 0, conn id: 2002, crypto map: CMAP
        sa timing: remaining key lifetime (k/sec): (4525504/3451)
        IV size: 16 bytes
        replay detection support: N
        Anti replay bitmap:
         0x00000000 0x0000001F
    outbound esp sas:
```



ISP router

Only three static routes:

50.0.0.0/24 via 20.0.0.1

60.0.0.0/24 via 20.0.1.1

100.100.100.0/24 via 20.0.2.1

Because of the double NAT, the ISP never sees a 192.168.x.x or 200.200.x.x address.

What broke

1. VPN traffic reached the ASA and got dropped. Packet Tracer wanted it permitted on the outside ACL.
2. The first ping always timed out. It was ARP, so I pinged the peer once before testing to cache ARP.
3. Object network group is not support on Packer Tracer ,so created objects for each identity NAT.

The files

GitHub: https://github.com/MoorthyRJ/Enterprise_pkt.git

Drive (.pkt file): https://drive.google.com/drive/folders/1dbFc0TPYua6pXBIzlctz5V4wbXUJLpQa?usp=drive_link

Open the .pkt in Packet Tracer 9.0.0 or newer.